Unit-2

Lab experiments

Experiment 1: Smart Fitness Watch Product Description by using google collab & python

Aim:

To design Zero-shot, One-shot, and Few-shot prompts to generate a product description for a Smart Fitness Watch.

Procedure:

1. Open Google Colab and create a new notebook.
2. Create three prompts using Zero-shot, One-shot, and Few-shot techniques.
3. Include product features and benefits in the prompts.
4. Display the prompts using Python's print() function.
5. Compare the prompts based on the instructions and examples provided.

Prompt:

1. Zero-shot Prompt:

“Write an attractive product description for a Smart Fitness Watch. Include its features, benefits, design, and target customers. Use simple and professional English.”

2. One-shot Prompt:

“Example: Product: Wireless Earbuds. Description: Enjoy clear sound, comfortable fitting, and wireless convenience for music and calls.

Now write a similar product description for a Smart Fitness Watch. Include its features, benefits, design, and target customers.”

3. Few-shot Prompt:

“Example 1: Product: Smart Bottle. Description: Tracks water intake and reminds users to drink water.

Example 2: Product: Fitness Band. Description: Tracks daily steps, calories, and physical activity.

Now write a product description for a Smart Fitness Watch. Include heart-rate monitoring, sleep tracking, step counting, notifications, benefits, and target customers.”

Result:

Three prompts were designed to generate a Smart Fitness Watch product description using Zero-shot, One-shot, and Few-shot prompting techniques.

Experiment 2: Blog on Applications of AI in Healthcare by using google collab & python

Aim:

To design Zero-shot, One-shot, and Few-shot prompts to generate a 200-word blog on "Applications of Artificial Intelligence in Healthcare."

Procedure:

1. Open Google Colab and create a code cell.
2. Prepare three prompts using different prompting techniques.
3. Include healthcare applications such as diagnosis, medical imaging, and patient monitoring.
4. Display the prompts using Python.
5. Compare the prompts based on their instructions and examples.

Prompt:

1. Zero-shot Prompt:

"Write a 200-word blog titled 'Applications of Artificial Intelligence in Healthcare'. Include disease diagnosis, medical imaging, drug discovery, patient monitoring, benefits, and challenges. Use simple English and a clear conclusion."

2. One-shot Prompt:

"Example: Title: AI in Education. Blog: Artificial Intelligence improves education through personalized learning, automated assessment, and intelligent tutoring systems.

Now write a 200-word blog titled 'Applications of Artificial Intelligence in Healthcare'. Discuss diagnosis, medical imaging, drug discovery, patient monitoring, benefits, and challenges. Follow the example's simple writing style."

3. Few-shot Prompt:

"Example 1: Topic: AI in Education. Style: Introduction, applications, benefits, and conclusion.

Example 2: Topic: AI in Agriculture. Style: Explain practical applications, advantages, and challenges in simple language.

Now write a 200-word blog on 'Applications of Artificial Intelligence in Healthcare'. Include an introduction, diagnosis, medical imaging, drug discovery, patient monitoring, benefits, limitations, and conclusion. Use clear and professional English."

Result:

Three prompts were designed to generate a healthcare blog of approximately 200 words using different prompting techniques.

Experiment 3: Article Summarization into 50 Words by using google collab & python

Aim:

To design Zero-shot, One-shot, and Few-shot prompts to summarize an article into 50 words and compare the summaries based on accuracy, completeness, and readability.

Procedure:

1. Open Google Colab and create a code cell.
2. Store the article in a Python string variable.
3. Prepare Zero-shot, One-shot, and Few-shot summarization prompts.
4. Instruct each prompt to produce a 50-word summary.
5. Compare the generated summaries based on accuracy, completeness, and readability.

Article:

"Artificial Intelligence is transforming healthcare by helping doctors analyze medical images, identify disease patterns, and support clinical decisions. AI systems can monitor patients and predict certain health risks. They may also accelerate drug discovery and reduce administrative work. However, healthcare AI requires reliable data, privacy protection, fairness, and human supervision."

Prompt:

1. Zero-shot Prompt:

"Summarize the following article in exactly 50 words. Preserve the main ideas, use clear language, and do not add information that is not present in the article."

2. One-shot Prompt:

“Example: Article: Online learning allows students to study remotely. It provides flexibility but requires internet access.

Summary: Online learning provides flexible remote education but depends on reliable internet access.

Now summarize the given healthcare article in exactly 50 words. Preserve its main ideas without adding new information.”

3. Few-shot Prompt:

“Example 1: Article: Solar energy produces electricity from sunlight. Summary: Solar energy converts sunlight into electricity.

Example 2: Article: Exercise improves physical fitness and health. Summary: Regular exercise improves fitness and supports health.

Now summarize the given healthcare article in exactly 50 words. Include its main applications, benefits, and limitations. Use clear, concise language and preserve the original meaning.”

Result:

Three summarization prompts were designed. Their generated outputs can be evaluated based on accuracy, completeness, readability, and word count.

Experiment 4: Professional Email Requesting Leave Due to Illness by using google collab & python

Aim:

To design Zero-shot, One-shot, and Few-shot prompts to generate a professional email requesting leave due to illness.

Procedure:

1. Open Google Colab and create a code cell.
2. Prepare three prompts with different levels of examples.
3. Include the subject, greeting, reason for leave, date, and closing.
4. Display the prompts using Python.
5. Compare the generated emails based on tone, grammar, formatting, and completeness.

Prompt:

1. Zero-shot Prompt:

“Write a professional email requesting leave due to illness. Include a subject line, greeting, reason for leave, requested leave date, polite request, and closing. Use formal and grammatically correct English.”

2. One-shot Prompt:

“Example:

Subject: Request for Permission

Dear Sir/Madam,

I request permission to attend an important personal appointment. Kindly consider my request.

Thank you for your understanding.

Regards, [Your Name]

Now write a professional email requesting leave due to illness. Include the subject, greeting, reason, leave date, and polite closing.”

3. Few-shot Prompt:

“Example 1: Purpose: Meeting request. Tone: Polite and professional. Format: Subject, greeting, message, and closing.

Example 2: Purpose: Assignment extension. Tone: Respectful and concise. Format: Subject, greeting, explanation, request, and closing.

Now write a professional email requesting leave due to illness. Mention [DATE] as the requested leave date, briefly explain the illness, request approval, and include a polite closing. Use proper email formatting.”

Result:

Three prompts were designed to generate professional illness leave emails. The outputs can be compared based on tone, grammar, formatting, and completeness.

Experiment 5: Promotional Social Media Post for an AI Workshop by using google collab & python

Aim:

To design Zero-shot, One-shot, and Few-shot prompts to generate a promotional social media post for an AI Workshop.

Procedure:

1. Open Google Colab and create a code cell.
2. Prepare three prompts for promotional content.
3. Include workshop benefits, practical sessions, and registration details.
4. Add instructions for emojis, hashtags, and an attractive headline.
5. Compare the generated posts based on creativity, relevance, and clarity.

Prompt:

1. Zero-shot Prompt:

“Create an attractive social media post promoting an AI Workshop. Mention workshop benefits, practical learning, and registration details. Use an engaging tone, suitable emojis, and relevant hashtags.”

2. One-shot Prompt:

“Example:

Python Workshop!

Learn programming through hands-on activities. Improve your coding skills and explore new opportunities.

Register now!

#Python #Workshop #Learning

Now create a similar promotional social media post for an AI Workshop. Mention AI applications, practical sessions, workshop benefits, and registration. Use emojis and relevant hashtags.”

3. Few-shot Prompt:

“Example 1: Style: Exciting headline, short sentences, emojis, and hashtags.

Example 2: Style: Professional language, benefits-focused content, and a clear call to action.

Now create a promotional social media post for an AI Workshop. Include an attractive headline, benefits of learning AI, hands-on sessions, date [DATE], venue [VENUE], registration details, and relevant hashtags. Use a friendly and professional tone suitable for LinkedIn or Instagram."

Result:

Three prompts were designed to generate promotional social media content for an AI Workshop using Zero-shot, One-shot, and Few-shot prompting techniques.